

Jaden Minnick

Blacksburg, VA & Harrisonburg, VA

jminnick22@vt.edu — +1-540-271-5045

minnick.vt.domains — linkedin.com/in/jaden-minnick

Education

Virginia Polytechnic Institute and State University | B.S. Chemical Engineering, B.S. Physics | Blacksburg, VA | *Expected May 2026*

Overall GPA: 3.72/4.0 | Minors: Nuclear Engineering, Mathematics

Technical University of Denmark (DTU) | Chemical and Biochemical Engineering | *Lyngby, Denmark* | GPA: 3.7/4.0

- Conducted laboratory experiments on Rotary Vacuum Drum Filtration, Batch Distillation, Liquid-Liquid Extraction, Fluidized Bed Drying, and Process Control.
- Prepared technical reports analyzing process efficiency, mass/heat transfer, similarities to theory, and operational challenges.
- Developed Python-based models to analyze thermodynamic and transport-intensive experiments, generate performance plots, and compare results to theoretical predictions.

Research Experience

Undergraduate Researcher | Virginia Tech Department of Nuclear and Particle Physics | *Blacksburg, VA* | Jan 2024 – Present

- Calibrated and characterized photomultiplier tubes for the Small Angle Monitors (SAMs), developing systematic routines for gain scans, single-photon response measurements, and stability assessment across varying high voltages.
- Engineered and optimized NIM crate configurations by integrating discriminators, amplifiers, attenuators, and fan-in/fan-out modules to ensure stable and reliable detector behavior.
- Performed scintillator and lightguide tests to evaluate energy deposition, collection efficiency, and light-tightness under experimental conditions.
- Developed Python/Linux workflows for data acquisition, digitizer processing, and automated QA/QC analysis, significantly improving reproducibility and detector performance evaluation.
- Presented calibration methodology, detector performance studies, and analysis procedures at multiple regional and national conferences, strengthening communication skills across interdisciplinary audiences.

Work Experience

Chemistry Recitation TA | Virginia Tech | *Blacksburg, VA* | Jan 2024 – May 2024

- Led weekly recitations and quizzes for 30+ students.
- Guided students through problem-solving and exam review.

Engineering Intern | MEI Engineering | *Harrisonburg, VA* | Jan 2022 – May 2022

- Designed circuits, lighting, and power layouts in AutoCAD/Fusion/DataCAD.
- Completed electrical load + distribution calculations.
- Coordinated with interdisciplinary teams on deadlines and requirements.

Medical Lab Assistant | Rockingham Memorial Hospital | *Harrisonburg, VA* | Aug 2021 – Jan 2022

- Supported Hematology, Chemistry, Microbiology, and Pathology operations.
- Performed QC, reagent management, and equipment maintenance.
- Troubleshoot diagnostic discrepancies with technicians and physicians.

Leadership Positions

Tutoring Committee Head | Alpha Chi Sigma | *Virginia Tech* | Sep 2023 – Present

- Volunteered over **400 hours** to help fellow Virginia tech students in their Chemistry, Physics, Chemical Engineering, Math, etc classes
- Oversaw fraternity academic support programs, expanded tutoring operations, and recruited mentors.
- Developed structured tutoring workflows, outcome tracking, and faculty collaboration.

Professional Committee Head | Alpha Chi Sigma | *Virginia Tech* | *Sep 2024 – Present*

- Organized industry talks, resume workshops, and networking events with alumni and professionals.
- Built partnerships with companies and VT departments to expand student career opportunities.

Vice President | Omega Chi Epsilon | *Virginia Tech* | *Jan 2025 – Present*

- Led academic programming, outreach events, and chapter operations.
- Coordinated faculty partnerships and student support initiatives.

Secretary | Omega Chi Epsilon | *Virginia Tech* | *Jan 2024 – Jan 2025*

- Managed documentation, communications, and meeting records.
- Supported logistics for workshops, academic sessions, and chapter events.

Engineering & Physics Representative | Office of Undergraduate Research | *Virginia Tech* | *Jan 2025 – Present*

- Acted as liaison between STEM departments and the undergraduate research office.
- Promoted funding opportunities and expanded student involvement in research programs.

Skills

- **Programming & Analysis:** Python (NumPy, SciPy, Matplotlib), ROOT, Matlab, Linux Bash
- **Modeling & Engineering Tools:** ANSYS (Fluent, Mechanical) Aspen Plus, SOLIDWORKS, Autodesk AutoCAD & Fusion, Mathematica
- **Instrumentation & DAQ:** NIM modules, PMTs, scintillators, digitizers, oscilloscopes, detector calibration workflows
- **Software:** L^AT_EX, HTML/CSS/JavaScript **Certifications:** Manufacturing Technician Certification (5s, Kaizen), Manufacturing Specialist Certification (6 sigma, lean), OSHA HazCom RTK Training, Cyber Security Awareness (CST001), Security Awareness Training (GEN034), Oxygen Deficiency Hazard (SAF103), Physics Division Work Governance (SAF116kd), General Access RWP (SAF801kd), Radiation Safety and Training, Introduction to Process Safety (ELA950), Hazard Recognition (ELA951), Identifying & Minimizing Process Safety Hazards (ELA952), Introduction to Managing Process Safety Hazards (ELA953), Inherently Safer Design (ELA984), Process Safety Lessons Taught by Experience (ELA908), Fundamentals of Process Intensification (ELA300), Intensified Reaction Processes (ELA301), Process Intensification in Separation Processes (ELA302), Fundamentals of Intensified Heat and Mass Transfer (ELA303), Intensified Distillation Processes (ELA304), Introduction to Modular Chemical Process Intensification (ELA352)

Presentations - Talks

Talks focused on detector calibration, diagnostics development, and engineering design.

- The Physics and Astronomy Congress (PhysCon), Denver, Colorado: Oct 2025
- Southeastern Section of the APS (SESAPS) Conference, James Madison University: Oct 2025
- Virginia Tech Chemical Engineering Showcase, Virginia Tech: Apr 2025
- Virginia Tech Chemical Engineering Seminar, Virginia Tech: Feb 2025
- Virginia Tech Chemical Engineering Showcase, Virginia Tech: Dec 2024
- American Physics Society DNP Conference, MIT: Oct 2024
- Virginia Tech Summer Research Conference, Virginia Tech: Jul 2024
- Virginia Tech Department of Physics Conference, Virginia Tech: Jul 2024

Affiliations

- **Alpha Chi Sigma** (AXΣ, Professional Chemistry Fraternity) (Dec 2023 – Present)
- **Omega Chi Epsilon** (ΩXE, **Chemical Engineering Honor Society**) (Dec 2023 – Present)
- **Sigma Pi Sigma** (ΣΠΣ, **Physics Honor Society**) (Apr 2025 – Present)
- **American Institute of Chemical Engineers (AIChE)** – Member (Sep 2022 – Present), Mentorship Program Participant (Sep 2023 – Present)
- **National Society of Professional Engineers (NSPE)** – Member (Aug 2022 – Present)
- **American Nuclear Society (ANS)** – Member (Aug 2022 – Present)

Engineering Projects

For more information please visit my portfolio website: minnick.vt.domains

Tokamak First Wall Cooling Channel (CFD, ANSYS Fluent), 2026

- Modeled conjugate heat transfer under 1–10 MW/m² heat flux.
- Conducted steady-state simulations with refined wall-fluid meshing.
- Performed parametric studies on coolant velocity and heat flux; analyzed thermal performance and safety limits.
- Verified accuracy via mesh convergence and sensitivity analyses.

Production-Scale Enzyme Processing System Design (Novonesis), 2026

- Designed an industrial-scale batch system for enzyme adsorption with integrated flow and dosing systems.
- Implemented automated feedback control for pH and temperature using real-time monitoring.
- Analyzed mixing, heat, and mass transfer to ensure scalability and consistent operation.

Reactor & Separation Process Optimization (Aspen Plus), 2025

- Designed reaction–distillation systems for acetonitrile, toluene, and benzene.
- Modeled kinetics, equilibria, and transport; sized columns and heat exchangers.
- Conducted energy and economic analyses; optimized cooling and heat integration.

Absorber Revamp & Heat Integration (Aspen Plus), 2025

- Redesigned reboiled absorber system to improve C₂/C₃ recovery.
- Optimized heat-exchanger configuration, identifying a two-cooler design.
- Reduced annual operating cost by 29% through heat-duty balancing.

Optical Pumping & Zeeman Spectroscopy (Rb-85/87), 2024

- Performed optical pumping and RF spectroscopy to measure hyperfine and Zeeman splitting.
- Aligned optical system and tuned magnetic fields for resonance detection.
- Analyzed absorption signals to identify isotopic transitions.

Hall Effect Semiconductor Characterization (Ge), 2024

- Measured Hall voltage using lock-in amplification under varying magnetic fields.
- Determined carrier type, concentration, and mobility from transport data.
- Applied semiconductor theory to interpret electronic properties.

Additional Projects

- Built Arduino-based ethylene sensor for fruit ripeness detection.
- Designed and tested 3D-printed wind turbine blades across wind conditions.